

Use-Case Flow Specification

Main Success Scenario

1. Actor navigates to the login page.
2. Actor enters their username (Member ID / Staff ID) and password.
3. System validates the credentials against the member/staff directory.
4. System creates a secure session and records the login timestamp.
5. System redirects the actor to their respective dashboard (Student or Librarian view).
6. Student Member searches the ISBN repository and selects a title currently marked “On Loan.”
7. System displays the title's details and a “Place Hold” option.
8. Student Member selects “Place Hold.”
9. System invokes «include» UC-06: Verify Membership Eligibility to confirm the member is not suspended and has not exceeded the maximum hold/loan limit.
10. System checks the member's outstanding overdue fines against the ₹500 threshold (FR-005).
11. System calculates the member's position in the existing FIFO waitlist for that ISBN.
12. System creates a new hold record, assigns the queue position, and displays a confirmation with the position number and estimated wait.
13. Use case ends successfully.

Alternate Flow — A1: Invalid Credentials

Branches from step 3, when the entered credentials do not match any account.

- 3a. System displays “Invalid username or password” and increments the failed-attempt counter for that account.
- 3b. If failed attempts reach 5 within 15 minutes, the system locks the account for 30 minutes and emails the actor a lockout notice.
- 3c. Use case ends without a session being created.

Alternate Flow — A2: Member Already Holding Maximum Allowed Reservations

Branches from step 4, when UC-06 determines the member already holds the maximum permitted number of active reservations (e.g., 3).

- 9a. System halts the hold-placement process.
- 9b. System displays a message: “You have reached the maximum of 3 active holds. Cancel an existing hold to continue.”
- 9c. Student Member may cancel an existing hold and retry, or exit.
- 9d. Use case ends without a new queue entry being created.

Alternate Flow — A3: Outstanding Fines Exceed Threshold

Branches from step 5, when total unpaid fines exceed ₹500 (FR-005).

- 10a. System halts the hold-placement process.
- 10b. System displays the outstanding balance and a “Pay Now” prompt.
- 10c. If the member pays the fine (see UC-05/UC-07 fine flows), control returns to step 5 for re-evaluation.
- 10d. Otherwise, the use case ends without a new queue entry being created.